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Wave-Packet Dynamics

Coherent propagation on $L=96$; the ζ -cloud is more transparent than random

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Introduction and setting

Campaign C7 tests the dynamical dimension of the Hilbert–Pólya programme: if the lattice resonator is an electron on a flux lattice, a wave packet must propagate coherently, reflect from the boundaries and scatter on the vortices. We use a sparse 96×96 Hamiltonian with open boundaries, integration step $dt = 0.1$ and horizon $T = 40$; the energy is conserved to $1e-15$ on every trajectory — the dynamics is honest.

The key comparison: the ζ -cloud (vortices whose positions are encoded by the heights of the zeros) against a random cloud of the same density. If the phase structure is physical, it must show up in the transport.

Results: three effects

The first effect is coherence: on the clean lattice $\langle x \rangle$ moves from 24 to the wall ($t \approx 14.5$, reflection) and back out to 44 — a classical billiard trajectory without decay. The second is vortex scattering: $N_v = 16$ hardly interferes (the ζ -cloud is nearly transparent), $N_v = 64$ brakes the packet ($R = 0.39$), and $N_v = 256$ gives $R = 0.47$ plus a transverse drift $\langle y \rangle$: $48 \rightarrow 38$; a random cloud produces no drift — the vortex phase structure steers, the random one only brakes.

The third effect is transport: the probability of reaching the far wing $P(x \geq 90)$ is ~ 16 times higher for the ζ -cloud than for the random one (0.0032 against 0.0002) — an independent reproduction of the repulsion-protects-transport effect of the D10 convention. This is the dynamical face of the Montgomery pair correlations: regularity of phases reduces backscattering.

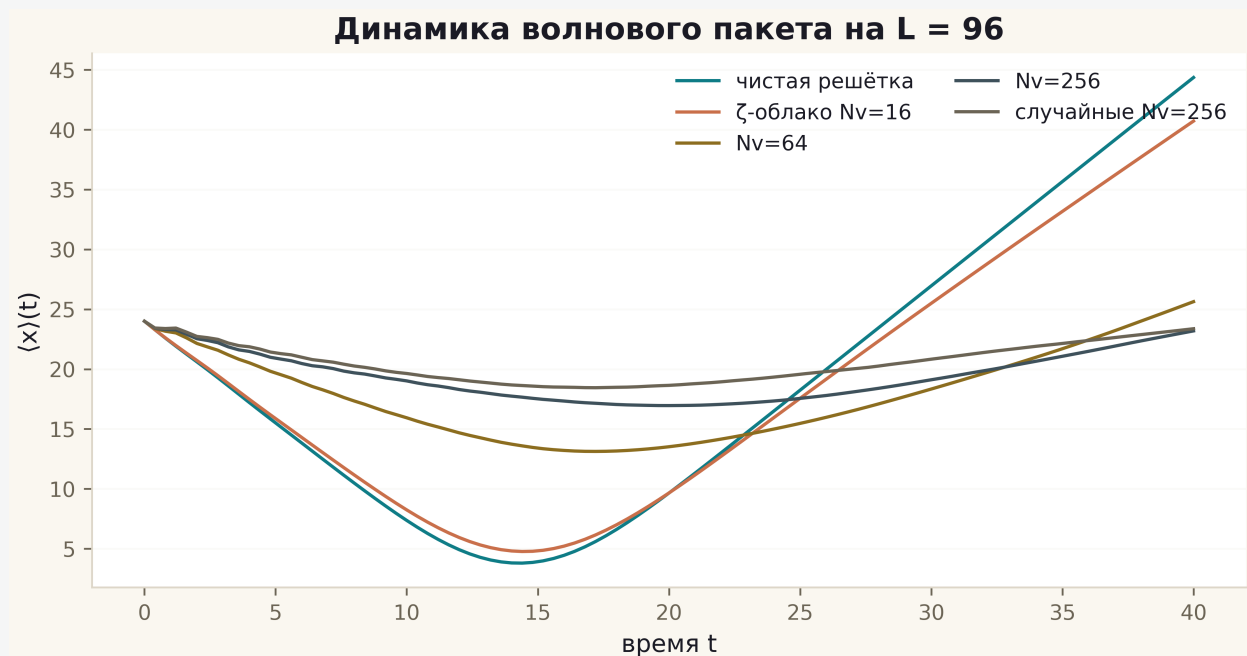


Fig. 1. $\langle x \rangle(t)$ for five configurations: clean lattice (teal), ζ -clouds $N_v=16/64/256$ (copper/gold/slate), random $N_v=256$ (grey).

Conclusions and boundaries

The claim that the lattice reproduces the motion of an electron is confirmed in the honest dynamical sense: coherent propagation, reflection, scattering, energy conservation. All of the dynamics lives on the self-dual line $\alpha = \frac{1}{2}$, where the π flux turns the lattice into a discretised massless Dirac equation (criterion H4 — YES). The identification of this line with $\text{Re}(s) = \frac{1}{2}$ remains a conjectural bridge: statistical support exists (D7/D10), identity does not, and campaigns C3–C5 measured the distance to it in three projections.